

Desert and Forest

*Kent Beck describes two working environments: the desert, open, fast, unobstructed, where you can move without friction, and the forest, dense, slower, where the ecosystem is richer and learning accumulates in the undergrowth.*¹

If the right structure holds understanding, what keeps it alive over time?

Beck built Extreme Programming² on the conviction that software development is a social activity, that the team is not overhead but the mechanism through which good software gets made. XP was largely set aside in favor of frameworks that separated thinking from building, planning from implementation, people from each other. This book is, in part, an argument that XP was right, and that AI is forcing us to rediscover why.

AI is the most powerful desert tool ever built. It removes almost every obstacle to individual speed, and a developer can move faster alone with an agent than any 10x engineer³ could move alone before. The desert has never been more accessible, or more tempting.

Which makes the forest more important than it's ever been, and more fragile.

The Trireme gave the structure. But the structure depends on a precondition it can't supply: a room where uncertainty is discussable.

The Agora chapter called psychological safety the prerequisite. The sharper version is *epistemic safety*: the conditions where people reveal what they don't know. Where the junior asks the question without calculating the risk first. Where the senior shares the reasoning, not just the conclusion. Where the half-formed thought gets voiced instead of suppressed. Where

¹The forest-and-desert metaphor is Beth Andres-Beck and Kent Beck's. See "Forest & Desert," *Tidy First?* (tidyfirst.substack.com/p/forest-and-desert), and Martin Fowler, "Forest And Desert" (martinfowler.com/bliki/ForestAndDesert.html).

²Kent Beck, *Extreme Programming Explained: Embrace Change* — discussed, with citation, in the introduction.

³"10x engineer" traces to Sackman, Erikson, and Grant's 1968 study on individual programmer productivity variance, popularized in software culture via Steve McConnell's *Code Complete*.

assumptions become visible before they become decisions.

The room needs the freedom to be uncertain out loud.

That room is the precondition for everything the book has argued. A team can't build shared understanding without it. A team can't get weighted validation without it. A team can't catch the XY problem without it. A team can't grow the next generation without it. A team can't have constructive friction without it.

The process creates the container; the safe room is what fills it.

What breaks the room is rarely dramatic. The subtler damage comes from well-intentioned behavior that lands wrong.

The engineer who questions everything in a review request. Rigorous, thorough, evidence-based. From the inside it feels like raising the standard. From the outside it feels like being interrogated. The junior stops raising concerns because last time it became a debate. The senior stops sharing uncertain thinking because uncertain thinking gets challenged. The room gets quieter. Safer-feeling on the surface. More fragile underneath.

Or the 10x engineer who is genuinely fast, genuinely capable, and experiences the team as an obstacle. Optimized for the desert, not malicious about it. They've been rewarded for individual speed their whole career. The Spec Session feels like overhead. The review process feels like a bottleneck. The team feels like it's slowing them down, because it is, because that's partly what the team is for.

I was somewhere on that spectrum. Not hoarding knowledge or dismissing others, but asking too many questions in the wrong format, ending messages with exclamation marks out of habit that read as aggression rather than enthusiasm, and once, memorably, sending colleagues links to blog posts explaining why direct messages destroy flow and that we should stick to processes.

I was right about the substance. Direct messages do interrupt flow. The blog posts probably made good arguments. But sending links to colleagues to explain why their behavior was wrong, however valid the point, reads as condescending. The message was about process. The impact was about respect. I knew as soon as I sent it. I apologized. The escalation came anyway.

The room was telling me the room was breaking, and not gently.

The ego that breaks the room isn't usually the ego that knows it's an ego.

It's the engineer who cares deeply, has strong opinions, asks hard questions, and doesn't yet

know how the questions land. The desert virtues, namely rigorous, fast, evidence-based, and process-oriented, are genuinely good engineering habits. They just work differently in the forest.

In the desert, questioning everything in a review request optimizes for correctness. In the forest, it optimizes for silence. The question that makes someone feel tested produces a room where people stop volunteering the uncomfortable truth. And the uncomfortable truth is exactly what the room needs.

AI accelerates this dynamic. The engineer already inclined toward the desert gets a tool that makes it so productive the team starts to feel like friction.

Previously, leverage meant individual output. The 10x engineer was the one who wrote code fastest, shipped most, closed the most tickets. That definition made sense when implementation was the bottleneck.

Until the spec was wrong. Until the direction was off. Until the edge case surfaced after the agent had run three times. Until the junior who would have caught it in the room had stopped speaking up.

When the agent handles implementation, the bottleneck moves. The constraint becomes the quality of the understanding before the agent runs: the spec, the shared context, the room where the direction got challenged before anyone built anything. The engineer who improves the quality of that room now has more leverage than the engineer who writes code fastest.

The 10x engineer of the AI era is the person who makes the room better. Who asks the question that surfaces the XY problem. Who creates the conditions where the junior speaks up. Who slows down the Spec Session just enough to catch the misunderstanding before it becomes rework.

A counterargument worth taking seriously: some of the best software ever written came from individuals or tiny teams. Linus Torvalds. Wozniak. The early internet. The desert can produce extraordinary things. What those examples don't prove is that the desert scales. They show what's possible when one person holds the full picture. The question is what happens when the system gets complex enough that no single person can hold it, when the architecture spans more context than one mind can carry, when the edge cases live in the gap between what one person knows and what their colleague knows. That's where the forest earns its place: by surviving what the desert can't.

I've developed a small habit at conferences and events. When standing in a group, never close the circle completely. Leave a gap. Physical space that says: this conversation isn't finished, you're welcome to join. No invitation required. Just an open space where someone could step in.

It's the same instinct as switching to English the moment I know there's someone nearby who isn't a German speaker. Not because they can't understand, often they can. But because engagement is easier when the barrier is lower. The appreciation shows immediately, even from people who would have followed in German. Something relaxes. The conversation opens.

It's about inviting collaboration before anyone has to ask for it.

The desert closes ranks: tight clusters, known quantities, optimized for the people already in the room. The forest leaves space before it's requested. It accepts the friction of the unexpected perspective because it understands that the unexpected perspective is often the one that matters.

The Spec Session works the same way. The session that starts with "we've thought about this and here's what we're doing" has closed the circle. The one that starts with "we have some thinking but we haven't decided" leaves the gap. The junior's edge case, the stakeholder's clarification, the concern nobody voiced in the corridor: those arrive through the gap. They don't arrive in a closed circle.

The gesture is small. The effect compounds.

Building the room is slower than moving alone. It requires attention to how things land, not just whether they're correct. It requires making thinking visible even when visible thinking feels inefficient.

It requires, sometimes, someone telling an engineer that their blog post links were not well received.

A team with a safe room and moderate individual capability will outperform a collection of 10x engineers who broke the room. Not because capability doesn't matter. It does. But because the room is where the capability compounds. Where the junior's edge case catches the senior's blind spot. Where the Spec Session surfaces the misunderstanding before the agent burns the budget. Where the different perspective changes the direction before it's too late to change.

Two things hold at any size. A team needs friction: a team where everyone always agrees is a mirror, and mirrors don't catch mistakes. And a team needs the room: without people saying what they actually think, nothing else holds.

The desert produces output. The forest produces learning. And the learning is the cost that stays quiet, which is why the desert always looks cheaper than it is.

The forest grows slowly but survives what the desert cannot.